GRPO unraveled: A Small-Scale Empirical Study of RLVR
CMPT 981 Project Proposal

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Abstract

Reinforcement learning with verifiable rewards (RLVR) together with Group Relative Policy Optimization (GRPO) has become the standard recipe for fine-tuning large language models (LLMs) on reasoning tasks with verifiable rewards. Recent theory provides concrete bounds on its dynamics like the success amplification property, the gradient gap, and the step-size threshold. However, they are derived for idealized scenarios that are impractical in LLM training. We propose a small-scale empirical study that tests these predictions on a synthetic task with a binary verifier. We vary initial success rate, group size, reward sparsity, KL strength, and step size, and compare each result with the corresponding theoretical prediction. We increase the difficulty of our task, starting with the bandit setting, then a synthetic task with multiple correct answers, and finally a small LLM.

1 Introduction and Motivation

Reinforcement learning (RL) learns a policy of actions from a reward signal that is often sparse and delayed. In the context of large language models (LLMs), RL is used to align the model's responses with human preferences or task-specific objectives, most notably through reinforcement learning from human feedback (RLHF) [Ouyang et al., 2022]. However, RLHF can be expensive and time-consuming, as it requires humans to label which responses they prefer, and those labels are used to train a reward model.

In 2025, DeepSeek-R1-Zero [Guo et al., 2025] showed that a language model could learn long chain-of-thought reasoning from a rule-based reward, mainly a binary check of whether the final answer is correct, without any human feedback. This approach is called reinforcement learning with verifiable rewards (RLVR). The model was trained using Group Relative Policy Optimization (GRPO) [Shao et al., 2024], which samples multiple answers per prompt and normalizes the advantages by the group standard deviation. Since then, practitioners have relied on rules of thumb, such as sampling more answers for problems with a low success rate, normalizing advantages by the group metrics, and averaging the loss over tokens. However, each of these choices changes the fundamental optimization problem, and it is still unclear which ones matter and why.

Initial theoretical work provides precise answers in simplified settings [Mroueh, 2025, Suk and Duan, 2025, Thrampoulidis et al., 2026, Bay and Yearick, 2026]. Examples include the success amplification property of GRPO [Mroueh, 2025], which states that, with a KL penalty, the success probability settles at a fixed point higher than that of the reference model, and the gradient gap [Suk and Duan, 2025], the difference between the expected score of correct and incorrect responses, which is used to derive a step-size threshold below which training converges. However, these results are derived under simplified conditions, far from the stochastic, finite-sample regime in which GRPO is actually run. For example, they assume exact expectations and, in the case of Mroueh [2025], an exactly solved

inner optimization problem. Testing them directly is also hard, because LLM experiments cannot isolate a single variable: the success rate, task difficulty, and response length all change together.

Our project aims to bridge the gap between theory and practice. We plan controlled experiments that will allow us to sweep a grid of parameters, including the initial success rate, group size, reward sparsity, KL strength, and step size. By changing one parameter at a time and observing the effect on the learning dynamics, we can test the theoretical predictions in a setting closer to practice, with finite samples, stochastic gradients, and a learned policy. These results will offer further insight into the dynamics of RLVR and GRPO in practical settings. This can lead to a better understanding of the optimization problem and help guide how these settings are chosen when training LLMs with verifiable rewards.

2 Related Work

GRPO and RLVR. GRPO [Shao et al., 2024] drops the value network of Proximal Policy Optimization (PPO). For each prompt, it samples a group of G answers and uses the group-normalized reward $(r_i - \bar{r})/\mathrm{std}(r)$ as the advantage. DeepSeek-R1 [Guo et al., 2025] scaled this up with rule-based verifiers, $G = 16$, and a small KL coefficient that keeps the policy close to a reference model. A later study, Dr. GRPO [Liu et al., 2025], argued that the std-normalization and the per-response length normalization introduce biases. RLOO [Ahmadian et al., 2024] instead uses a leave-one-out baseline, the mean reward of the other $G - 1$ answers, which reduces variance without biasing the gradient. We use all three as advantage variants in our experiments to see how they affect the learning dynamics.

Success amplification [Mroueh, 2025]. In 2025, Mroueh analyzed GRPO in the RLVR setting and identified its success amplification property. The paper shows that, for binary rewards with a reverse-KL penalty and no clipping, GRPO's objective can be seen as a contrastive loss between correct and incorrect samples taken from the previous policy. Solving the KL-regularized problem exactly at each iteration shows that the success probability follows a recurrence $p_n = h(p_{n-1})$ for an explicit function h. Every fixed point of this recurrence satisfies $p^* > p_{\mathrm{ref}}$, which means that GRPO increases the success probability above that of the reference model. In this project, we will test whether SGD training reaches the predicted p^* for different values of β (the strength of the KL penalty) and for finite G. We will also compare this reverse-KL penalty, $\mathrm{KL}(\pi_\theta \,\|\, \pi_{\mathrm{ref}})$, with the forward-KL penalty $\mathrm{KL}(\pi_{\mathrm{ref}} \,\|\, \pi_\theta)$, which lets us distinguish mode-seeking (reverse) and mass-covering (forward) regularization.

Gradient gap and step-size thresholds [Suk and Duan, 2025]. The paper introduces the gradient gap, defined as the difference between the expected score of correct and incorrect responses. It proves a sufficient step-size condition for convergence and proves lower bounds where a constant step size makes performance drop to zero. To focus purely on optimization dynamics the authors first validate their framework on controlled bandit problems. With exact gradients, the bound scales as $1/T$ in the response length T and shrinks with the success rate through $(1 - p)$. The paper argues that GRPO reaches this scaling through its normalization. The authors prove that policy improvement in RLVR follows a logistic growth relation with the cumulative alignment gap. They further show this in simulation and in actual LLM post-training. We will verify these claims in a stochastic setting and see how the threshold moves when responses have different lengths but share the learning rate.

Group size and informative prompts. A group with all-equal rewards has zero advantage, and a common practice is to filter out such groups. Bay and Yearick [2026] show that GRPO and Dr. GRPO differ only in how they use the group standard deviation. They also derive a closed-form group-size law in $G\,p(1-p)$, and check it on a Bernoulli toy model by subsampling rollouts. Nie et al. [2026] prove that when the success rate varies across prompts, the fraction of all-equal groups exceeds the prediction from the average success rate, and that a fixed-reference advantage $A = 2r - 1$ outperforms Dr. GRPO on GSM8K. Plyusov et al. [2026] show that a group can have mixed rewards yet miss rare correct solutions, with a non-monotonic dependence on G. We plan a controlled study of the group-size law, where we can set the success rate and group size independently. This will allow us to test the law's predictions and see where it breaks down.

Pass@K advantage shaping. Thrampoulidis et al. [2026] show how advantage shaping corresponds to maximizing surrogate rewards and derive an unbiased leave-one-out pass@K estimator.

Their population analysis explains how different advantage rules affect optimization. They also present initial experiments and identify finite-sample effects as a direction for further study. We build on this work using controlled tasks where the success probability can be computed exactly, allowing us to test how closely training with different group sizes follows the population predictions.

Entropy and diversity collapse. Yue et al. [2025] show that RLVR-trained models beat their base model at small k but lose to it at large k, which suggests that RLVR mostly sharpens what the base model can already do.

3 Problem Formulation

We start with a prompt x and a policy $\pi_\theta(y \mid x)$ that generates a response y. A verifier returns a reward $r(x, y) \in \{0, 1\}$, and the policy succeeds on prompt x with probability $p(x)$. We sample a group of G responses per prompt, compute advantages A_i within the group, and take a gradient step to increase

$$J(\theta) = \mathbb{E}[r] - \beta \, \mathrm{KL}(\pi_\theta \,\|\, \pi_{\mathrm{ref}}). \tag{1}$$

Our search space consists of the following variables: the initial success rate p_0; the group size G; the reward sparsity, set by the answer length L and vocabulary size K (a uniform policy succeeds with probability K^{-L}, so we set p_0 separately with a short supervised warm-start); the KL strength β; the step size η; and the advantage type (GRPO, mean-only, RLOO, or pass@k).

We use three policies of increasing realism. *Bandit:* $L = 1$, with a separate softmax over the K options for each prompt, so the theory's assumptions hold except for finite G and stochastic gradients. *Synthetic task:* $L \geq 1$, and the policy is a product of L softmaxes computed by one network shared across prompts. *Small LLM:* a small pretrained LLM trained on GSM8K [Cobbe et al., 2021], a real-world example in which $p(x)$ is unknown and must be estimated by sampling.

4 Plan

Our main goal is to test the limits of the theoretical predictions mentioned in the related work. Those works assume simplified conditions; we will test where the predictions break when those conditions are not fulfilled.

We will run a grid of experiments over the variables defined in Section 3. Ideally we will cover the entire grid, but we prioritize finding where the theoretical predictions break, and then sample the grid more densely around those limits.

We define the following core experiments, in order:

E1 Initial success rate $\times$ group size. Our first test checks the effects of the initial success rate and the group size on the learning speed. We will also compare our measured learning speeds with the group-size law of Bay and Yearick [2026].

E2 Success amplification [Mroueh, 2025]. In the simpler bandit setting ($L = 1$), we will test the success amplification property of GRPO by comparing the fixed points measured at finite group size with the infinite-group fixed points of Mroueh [2025]. We then repeat the measurement with the forward-KL penalty, for which the paper gives no prediction.

E3 Step-size threshold [Suk and Duan, 2025]. We will test the step-size threshold on the synthetic task. We will sweep the learning rate over a log grid and look for a narrow range where success first rises and then stalls or collapses. We will also check whether the threshold shrinks roughly as $1/L$ and as p_0 grows.

E4 Pass@k and response diversity [Thrampoulidis et al., 2026]. On a synthetic task with multiple correct answers, we will compare GRPO with biased and unbiased pass@k estimators across group sizes, measuring pass@1, pass@k, and response diversity under a fixed sampling budget.

E5 Escaping very sparse rewards. When rewards are sparse and almost no group has mixed rewards, training stalls. We will test what helps the model escape this plateau.

E6 Small-LLM check. We will train a small LLM to check whether the theoretical predictions and the insights from the previous experiments carry over to a real model.

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